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Evaluation of ICT for Networked Control Systems of Latency-Critical Applications in Production

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Abstract

This paper evaluates information and communication technologies (ICT) for networked control systems (NCS) of latency-critical applications in production. Latency-critical applications in production (e.g. closed-loop control or motion control) are focused when optimizing production. However, data capturing of complex latency-critical applications is a challenge. Wired solutions hamper its flexibility. Wireless technologies, however, might lack capability to process data with required latencies and reliability.

First, a morphology to characterize latency-critical applications is presented. Second, current ICT (e.g. 5G, LTE, WiFi) are typified. Third, ICT are clustered regarding their process and business potential to control latency-critical applications in production. Finally, the approach is validated with the use cases motion control, factory automation and automated guided vehicles.

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Keywords: Latency-Critical Applications; Networked Control Systems; ICT; 5G-technology; ICT; Smart Manufacturing.

1. Introduction

Challenges for production companies are constantly rising: While product life cycles are decreasing, market requests higher customization and quality at the same [1]. To comply with these rising challenges, companies constantly must improve their value creation by e.g., increasing quality and flexibility of production. For the last two decades, automation and digitalization are major enablers for described value creation [2]. Use cases like closed-loop process control, motion control, cloud motion control of automated guided vehicles or mobile robot control promise a great market potential and are focus of current research [3].

However, these use cases are rarely implemented in industry yet. The main reason for this are high application requirements regarding maximum application-perceived latency on the control level, especially of closed-loop control systems [4]. To implement beforementioned use cases for applications such as printing machines, presses or machine tools maximum

latencies of control systems between 0.25 ms and 50 ms must be realized [5]. Nowadays, closed-loop control is often realized through wired communication [6]. However, when it comes to realizing the mentioned use cases and to tackle the rising challenges, wired communication will reach its limits [7]. Therefore, wired communication must be replaced by wireless alternatives. Wireless communication has several technical advantages, e.g. enabling higher flexibility while mitigating cable breakage. Hard to reach locations or remote locations become more easily accessible in wireless communication systems. These technical advantages of wireless information and communication technologies (ICT) result in cost reduction [8, 9].

Main disadvantage of current wireless ICT, especially for latency-critical application, are both minimal latencies and reliability of ICT. The fifth generation (5G) of mobile network and its ability of ultra-reliable low-latency communication (URLLC) appears to be major enabler when it comes to improving production.

Until 2030, 5G promises to increase worldwide production GDP by \$740bn [10]. Nevertheless, not all use case probably require 5G. For many cases existing and cheaper alternatives of wireless communication could already bring major benefits for the production system.

Therefore, this paper evaluates ICT for networked control systems (NCS) in production. *Chapter 2* summarizes important terms and definitions. *Chapter 3* describes ICT requirements of NCS of latency-critical applications in production and develops a morphological box. *Chapter 4* evaluates ICT based on morphological box. *Chapter 5* shows applicability of morphological box based on three use cases.

2. Terms & Definitions

2.1. Latency-Critical Application

Machines and powered tools carrying out discrete production processes are summarized as *applications*. When carrying out discrete production processes, applications perform various actions. Each action is triggered by an event. The time between event and action of is defined as *application-perceived latency*, as *Fig. 1* shows [11].

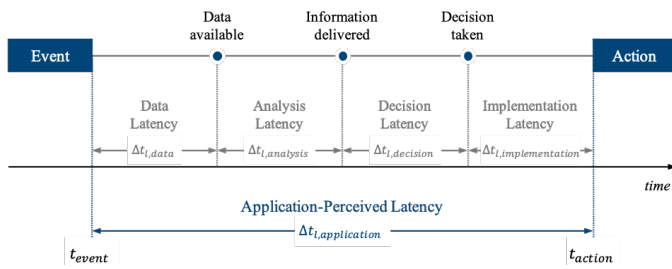


Fig. 1. Composition of application-perceived latency of applications in production.

In literature, the term *latency-critical* is often used synonymously to the terms *time-critical* and *real-time*. To form a clear understanding within this paper, the term *latency-critical application* is defined based on the definition of a hard real-time tasks. A real-time task is defined as hard if executing the action after its deadline causes damage on the system or even leads to a system breakdown [12]. Thus, latency-critical applications have a deadline $t_{action, deadline}$ defining this latest point in time at which application must execute its specified action without causing failures. The action deadline is related to the point in time of the event t_{event} . *Eq. (1)* shows formal requirement (R) of a latency-critical application, which is called *timeliness* [13].

$$R \equiv t_{event} + \Delta t_{l,application,max} \leq t_{action,deadline} \quad (1)$$

Based on the timeliness, *Eq. (2)* defines maximum application-perceived latency an application can allow.

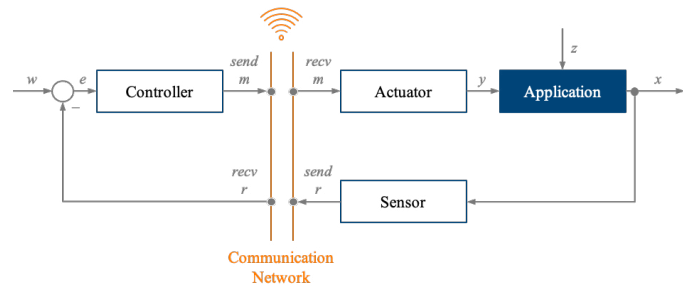
$$\Delta t_{application,max} \leq t_{action,deadline} - t_{event} \quad (2)$$

Latency-critical application is thus characterized by maximum application-perceived latency $\Delta t_{application,max}$, which must not be exceeded to ensure application's functionality.

2.2. Networked Control Systems

As describe in *chapter 1*, wireless communication has several technical advantages for production systems and is key enabler of future use cases. Therefore, wireless closed-loop control system, so called *networked control systems* (NCS), have been and still are main research focus in academia as well as in industry [14]. The general working principle and components of NCS are the same as for closed-loop control systems. However, controller, actuators and sensors are interconnected by a communication network, as *Fig. 2* shows. This network can also be shared with other control loops. Different architectures of NCS are existing. In one NCS, all components can be time-driven, event-driven or a mixture of both. This depends on the control techniques used [15, 16].

Fig. 2. Architecture of a networked control system.



2.3. Latencies in Networked Control Systems

To be applicable for latency-critical applications in production, NCS must fulfil timeliness requirement (*Eq. (1)*). Application-perceived latency in an NCS consists of its components' latencies, as *Fig. 3* demonstrates.

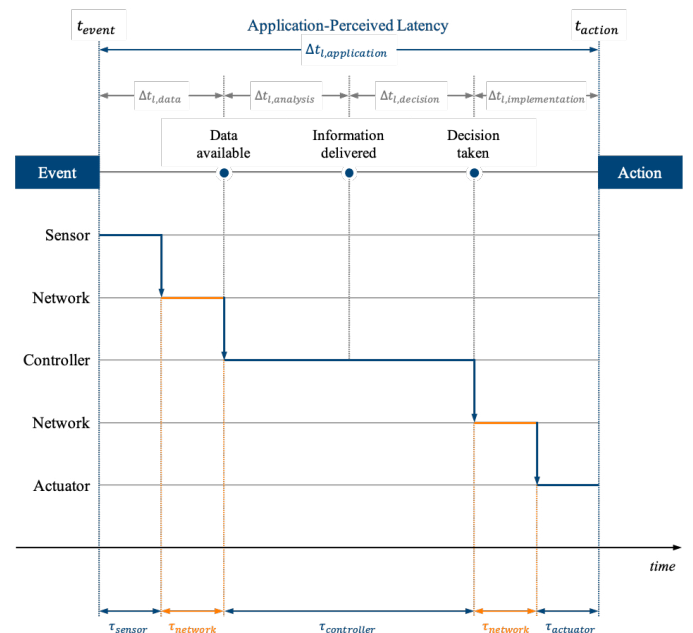


Fig. 3. Latencies in a networked control system.

Sensor latency τ_{sensor} describes the time in-between an event is sensed and recorded. Controller latency $\tau_{\text{controller}}$ expresses the time the controller requires to analyze data from sensor and decide on a control action, which is fed back to the actuator. Actuator latency τ_{actuator} describes the time in-between the reception of the feedback variable and the start of the physical execution of the action. Network latency τ_{network} describes the time the network requires to transmit data from sensor to controller and from controller to actuator. Network latency therefore occurs twice in an NCS. Eq. (3) shows composition of application-perceived latency in NCS:

$$\Delta\tau_{l,\text{application}} = \tau_{\text{sensor}} + \tau_{\text{controller}} + \tau_{\text{actuator}} + 2 \times \tau_{\text{network}} \quad (3)$$

Compared to application-perceived latency of wired control loops, network latency is the only additional latency. Wires also have a small latency, but it is less than one microsecond and therefore negligible for latency-critical applications in production. Network latency is thus decisive when it comes to the decision, whether NCS is applicable for latency-critical application. Eq. (4) defines maximum network latency $\tau_{\text{network,max}}$ of NCS:

$$\tau_{\text{network,max}} \leq \frac{\Delta\tau_{l,\text{application,max}} - \tau_{\text{sensor}} - \tau_{\text{controller}} - \tau_{\text{actuator}}}{2} \quad (4)$$

Whether NCS can fulfill this requirement mainly depends on information and communication technology (ICT), which will be further defined in chapter 2.4.

2.4. ICT

Information and communication technologies (ICTs) are “technologies used by people and organizations for their information processing and communication purposes” [17]. ICTs find applications in different contexts and thus lead to different dimensions of definitions. They include all communication and related technology that provide access to information. Examples are wireless networks, the internet or cell phones. ICTs as part of digital communication systems become increasingly essential to an organization’s way of creating value. For smart manufacturing for example, ICTs utilizing networked data and information are used in order to govern manufacturing operations in the framework of Industrie 4.0 [18].

Wollschlaeger et. al. investigate the development of industrial communication. First automation networks were fieldbus systems that enabled simplified communication field and control-level of production. Afterwards Ethernet-based networks dominated communication in industrial automation. Nowadays, development of wireless networks is important, especially regarding the pursued advantages of wireless connection in production industry [19].

As the focus on this paper lies on NCS, further course of this paper only considers wireless ICT.

3. ICT-Requirements of NCS for Latency-Critical Applications in Production

To implement NCS for latency-critical applications in production, suitable ICT is key, especially to realize $\tau_{\text{network,max}}$. However, besides latency, other requirements must be considered choosing ICT. The following chapter presents decision characteristics as well as their configurations. Fig. 4 summarizes the results.

Latency - ms

(End-to-end) Latency is the time that it takes to transfer application data of a given size from a source to a destination, from the moment it is transmitted by the source to the moment it is successfully received at the destination (one-way latency) [20]. In other words, the end-to-end (E2E) latency, typically expressed in ms, equals τ_{network} and is the main decision criteria when it comes to choosing the right ICT for the NCS. The unit of latency in this context is millisecond. The configuration of the latency ranges from >1s to <1 ms.

Reliability - %

The required (service) reliability of the ICT used in the latency-critical application can be understood as the ability to perform as planned and required for a given time interval under predefined and given circumstances without failure [21]. It is therefore expressed as the probability that a data of size D is successfully transferred within a time period T [22].

Availability - %

The communication service availability relates to the ability to allow correct operation of the application. It is defined as the “percentage value of the amount of time the end-to-end communication service is delivered according to an agreed Quality of Service, divided by the amount of time the system is expected to deliver the end-to-end service according to the specification in a specific area” [20].

User Data Rate - Mbps

User data rate defines the minimum value of the number of bits transmitted or received over time, typically expressed in Mbps. This means, if the minimal required data rate is not achieved, execution of action will be of insufficient quality [23].

Connection Density - Dev./m²

Required connection density is measured in devices/m². It is defined as the number of devices performing on a certain area unit. The characteristic can be specified regarding the maximum, minimum or average value. It is also defined per volume, depending on the given application and use case. Because of the ICT’s variety in this characteristic, for certain use cases special requirements are given. Hence, only some of the possible ICTs are appropriate and cost efficient [24].

Localization Precision - m

Positioning is important to ensure correct tracking of devices, guarantee safety standards and enable applications in factory and process automation. For some use cases an accuracy within a few centimeters (cm) is required, whereas other applications need a positioning of some meter (m). For most control and automation systems a positioning precision in sub-meter accuracy is efficient. [25].

Communication Range - m

The communication range in meter indicates the possible range a communication technology can enable the transfer of data and information between sender and receiver [26].

Mobility – km/h

Industrial use cases with requirements regarding the communication technology can be ranked regarding mobility. Some applications are characterized by a higher mobility, for example in transportation or logistics, whereas others are more local and ranked with low mobility. Configurations range from < 5km/h to 500km/h [27].

Morphology in Fig. 4 summarizes the results. Thereby, configurations are ordered according to the increase in requirements from left to right. This means that the further to the right the configuration of the respective characteristic is, the higher the requirements for the ICTs.

Characteristic	Configurations						
Latency [ms]	>100	100–50	50–20	20–10	10–1	<1	
Reliability [%]	<90	90–99	99–99.9	99.9–99.99	99.99–99.999	99.999–99.9999	>99.9999
Availability [%]	<90	90–99	99–99.9	99.9–99.99	99.99–99.999	99.999–99.9999	>99.9999
Data Rate [Mbps]	<5	5–100	100–1000	1000–5000	5000–10000	>10000	
Connection Density [Dev./m ²]	<0.01	0.01–0.1	0.1–1	1–10	10–20	>20	
Localization Precision [m]	N/A	>10	10–1	1–0.1	0.1–0.01	<0.01	
Communication Range [m]	<1	1–5	5–10	10–50	50–100	100–1000	>1000
Mobility [km/h]	N/A	<5	5–50	50–100	100–500	>500	

Increasing Requirements →

Fig. 4. Morphological box of ICT's characteristics and their possible configurations for NCS.

4. Evaluation of ICT for NCS of Latency-Critical Applications in Production

On basis of morphology, different wireless ICTs are compared. Aijaz and Sooriyabandara consider possible ICTs enabling latency-critical applications in the industrial internet [28]. Based on their preselection, as well as the analysis of Wollschläger et. al, ten ICT are evaluated with the developed morphology, namely 4G, 5G, 6LoWPAN, Bluetooth, ISA 100.11 LoRaWAN, Wi-Fi 6, WirelessHART, WISA and ZigBee.

4G and 5G are the 4th respectively 5th generation of mobile communication standard. The 3rd Generation Partnership Project (3GPP) is responsible for standardization and specifications.

Wireless Local Area Network (WLAN/ Wi-Fi) stands for a standard for local computer networks according to IEEE 802.11 standard, which was first introduced in 1997. The sixth generation of WLAN, Wi-Fi 6 (IEEE 802.11ax) is currently in development. It aims at increased coverage, capacity and efficiency and is thus a choice for IoT applications [29]. A worldwide interoperability certification is planned for early 2021 by the Wi-Fi Alliance members [30].

IEEE 802.15 defines standards for wireless personal area networks (WPAN). Most common for industrial applications are IEEE 802.15.1 and IEEE 802.15.4 [28]. IEEE 802.15.1 serves as standard for WISA and Bluetooth. Based on IEEE 802.15.4 standard, many network variants provided: Most common is the transmission in the 2.4GHz range with a gross data rate of 250kbit/s. This includes ZigBee, ISA, WirelessHART and 6LoWPAN

Low Power Wide Area Networks (LPWAN) are radio networks specially designed for low power consumption, small data volumes and transmission over long distances or within buildings. The most common network is the Long Range Wide Area Network (LoRaWAN) LoRaWAN specification is currently worked out by the LoRa Alliance in order to standardize LPWAN [31].

Fig. 5 evaluates the ten ICT based on the morphology. Not every ICT is standardized respectively defined for each characteristic. In these cases, configurations are left blank for the respective ICT.

Characteristic	Configurations						
Latency [ms]	>100	100–50	50–20	20–10	10–1	<1	
Reliability [%]	<90	90–99	99–99.9	99.9–99.99	99.99–99.999	99.999–99.9999	>99.9999
Availability [%]	<90	90–99	99–99.9	99.9–99.99	99.99–99.999	99.999–99.9999	>99.9999
Data Rate [Mbps]	<5	5–100	100–1000	1000–5000	5000–10000	>10000	
Connection Density [Dev./m ²]	<0.01	0.01–0.1	0.1–1	1–10	10–20	>20	
Localization Precision [m]	N/A	>10	10–1	1–0.1	0.1–0.01	<0.01	
Communication Range [m]	<1	1–5	5–10	10–50	50–100	100–1000	>1000
Mobility [km/h]	N/A	<5	5–50	50–100	100–500	>500	

4G 5G 6LoWPAN Bluetooth ISA 100.11
 LoRaWAN Wi-Fi WirelessHART WISA ZigBee

Fig. 5. Evaluation of industrial ICT regarding the defined requirements of NCS for latency-critical applications in production.

5. Application of Morphology: Motion Control

To test and validate the applicability of the developed morphology, three use cases that are stated as latency-critical in literature were applied, namely motion control, factory automation and automated guided vehicles (AGV).

Motion control is a process in which moving and/or rotating parts of a machine are controlled. As method of process control it supports applications in different contexts, for example in industry robots. For instance, it supports in controlling printing presses, machine tools or packaging machines [32].

Factory automation, also referred to as discrete automation, include automated applications that are characterized by real-time control of machines and devices. Discrete automation is often used in manufacturing lines and fast production environments. Automated control, optimization and monitoring using closed loop systems are solutions that serve for robotics and computer-integrated manufacturing. Here, machine parts are in motion on a restricted space [33].

AGVs are unmanned vehicles with an own traction drive that is automatically controlled and guided without contact. AGVs are used for material transport, namely for pulling or carrying conveyed material with active or passive load suspension devices.

Fig. 6 classifies the described use cases are classified regarding their ICT requirements. If the dot for the ICT is right from the dot of the respective use case, the ICT is suitable for the use case.

The application of the morphology for the motion control use case shows, that 5G-technology is the only suitable ICT due to high requirements regarding latencies, reliability, and availability. For the other characteristics, alternative ICT are applicable. For factory automation and AGV however, other ICTs might fulfill the technical requirements as well. AGVs for example might be realized by Wi-Fi as well.

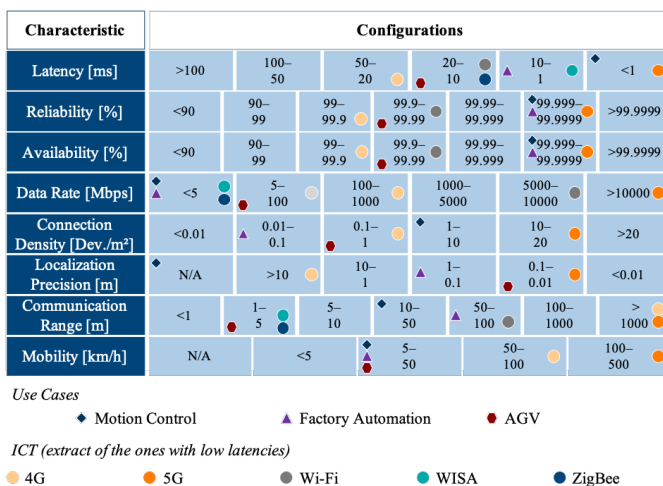


Fig. 6. Application of three latency-critical use cases to the developed morphology.

6. Summary & Outlook

The paper developed a morphology to support the choice of a technically suitable ICT for NCS of latency-critical applications in production. First, important decision characteristics were identified and explained. Second, ICT were classified regarding these requirements and possible configurations. Third, morphology was tested with three latency-critical use cases to show applicability. This application showed that morphology in general helps to classify possible ICT based on technical parameter. However, to sufficiently support decision process, economic evaluation of each ICT needs to be conducted. Especially in case that more than one ICT might be a suitable option, the influence of ICT on the latency critical application must be analyzed, meaning how many errors really occur due to not fulfilling maximum application-perceived latency as defined in Eq. (2).

Especially for the 5G-technology, this evaluation is important. First research has been conducted[34, 35]. These approaches further need to be developed, especially regarding the data-based approach on the control level.

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